Main Wheel-Skis (Mod S.O.O. 6001 and Mod 6/2182) – General data

1. General

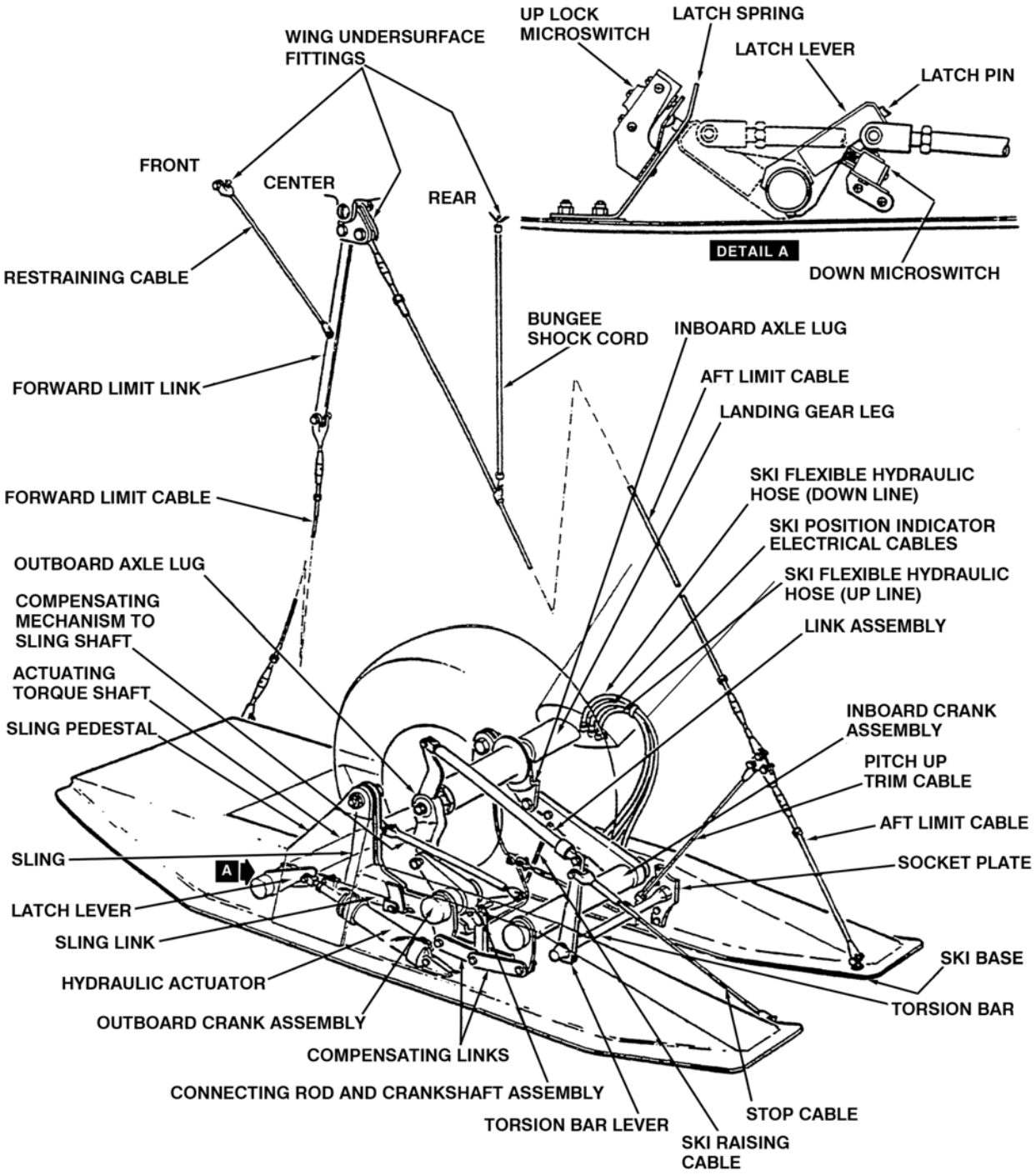
The left and right-hand Main Wheel-Skis form part of the Wheel-Ski installation (refer to 32-00-00 Landing Gear – General Data). Each ski is connected to its appropriate main leg axle ski attachments by its inboard and outboard crank assembly levers, and by cables, link and bungee cord to three fittings on the lower surface of the wing, directly above each ski inboard beam. The skis, controlled by a lever in the flight compartment, are hydraulically actuated from the aircraft's hydraulic system to the "DOWN/SNOW" or "UP/LAND" positions (refer to 32-30-00, Extension and Retraction (Wheel-Ski) (Mod S.O.O. 6001) – General Data). The skis are rigged level in flight and are self-trimming in pitch for either the "DOWN/SNOW" or "UP/LAND" configurations. Both skis embody a device for controlling the lateral attitude of the skis, regardless of the amount of main gear leg deflection, throughout its range of travel from in-flight to maximum landing load conditions. This feature applies only when the skis are in the "DOWN/SNOW" position. A selection to the "UP/LAND" position cancels this feature and introduces a device for re-trimming the lateral attitude for optimum ground clearance during taxiing. An indication of ski position is provided, on the same panel incorporating the Wheel-Ski selector lever.

The Main Wheel-ski base, (Mod 6/2199) consists of a UHMW layer retained by a series of steel retainer strips, screws and fittings. The UHMW base is trimmed out around fittings where tension cables are attached. Fasteners run through the retainer strips bolting the base to the ski. The base material provides a low friction surface on the bottom of the ski and helps protect the skis from damage experienced during landings. The retainer strips are wear components and should be inspected in accordance with PSM 1–6–7 and replaced as required.

2. Description

Refer to Figure 1 for details.

Each main Wheel-Ski consists of two beams structurally joined at the forward end and connected at the rear by the inboard crank assembly tube. A double acting hydraulic actuator is housed in the outboard beam of the ski, and is connected by its cylinder end to the beam structure. The rod end of the actuator connects to a latch lever to operate the ski sling, through actuating torque shaft levers and sling links. A latch pin on the latch lever penetrates the latch to retain the up lock when hydraulic power is lost. Hydraulic lines to the actuator are routed from connections on the inner side of the ski inboard beam, through the beam and the ski front structure. Two pedestals, one on each ski beam, form the attachment and pivot points for the ski sling. The inboard and outboard crank assemblies, which connect to the wheel axle inboard and outboard lugs, are interconnected within the outboard beam by two sets of links. The center of the two sets of links is operated by a connecting rod and crankshaft assembly, which in turn is operated by a shaft connected to the ski sling. This mechanism allows freedom of vertical and pitch motion, but controls the lateral angle of the ski, which is changed from level in the ski down configuration to an anhedral attitude when retracted. A torsion bar, mounted transversely between the beams at the rear of the ski, is secured in a socket plate attached to the inboard ski beam, and pivots in a pivot bearing attached to the outboard ski beam. A lever, secured to the outboard end of the torsion bar is connected to the upper arm of the outboard wheel axle lug by a link assembly, and to an eye bolt at the trailing edge of the outboard ski beam by a pre-tensioning cable. A cable interconnecting the ski sling actuating mechanism and the inboard crank assembly lever, provides the means of raising the ski for a "wheel - landing". A pitch down limit link and cable is connected between an eye bolt at the front of the ski and a center fitting on the undersurface of the wing. Two cables, connected together, are connected between an eye bolt at the rear of the ski, and the center fitting on the undersurface of the wing, to limit the pitch up angle of the ski in flight. A cable interconnecting the aft limit cable is routed through a pulley and fairlead to the ski sling. Two limit microswitches are installed within the outboard ski beam, both operated by the latch lever in its extremes of travel, to provide ski position indication.



Main Wheel-Ski
Figure 1

3. Operation

When skis “DOWN/SNOW” is selected, the actuator retracts, withdrawing the latch-pin from the latch spring, unlocking the latch lever and, through the connecting linkage, moves the sling down under the wheel, in doing so the sling impinges on the tire and forces the wheel to roll over the top of it so moving the ski downward into the extended position. Simultaneously, as the sling moves rearward, the ski raising cable, connected between the inboard crank assembly lever and the sling actuating linkage, is slackened progressively with sling movement. On selecting “UP/LAND”, the actuator extends moving the sling up and forward, allowing the wheel to roll from it, at the same time engaging the latch, simultaneously the actuating linkage operates the ski raising cable, which lifts the ski. Pitch down angle of the ski is restricted by the limit link and cable. The restoring force required to counteract external pitch down loads is provided by the pre-loaded torsion bar, geometrically linked to the landing gear leg, to yield equal forces in both the “UP/LAND” and “DOWN/SNOW” configurations, regardless of landing gear leg deflection. The pitch up angle of the ski in flight is fixed by the cables connecting the aft end of the ski to the center undersurface wing fitting. The restoring force, required to counteract external pitch up loads while taxiing on wheels, or skis, is provided by the spring loaded link connected between the torsion bar lever and the upper arm of the outboard wheel axle lug. To compensate for the change in levels between the “DOWN/SNOW” and “UP/LAND” positions of the ski, the trim cable interconnecting the two aft limit cables and the ski sling takes up slack in the aft limit cables during the transition between positions. The inboard and outboard crank assembly levers, which are of a different length, compensate for landing gear and tire deflection.

4. Servicing

A. Main Wheel-Ski Lubrication

Note The lubrication points, method, and frequency of application are given in the Main Wheel-Ski Lubrication Diagram. (Refer to 12-20-10, Lubrication – General Data, Figure 2).

5. Removal/Installation

Refer to Figure 2 for details.

A. Remove Main Wheel-Skis

Note The following removal procedures are based on the assumption that the aircraft is in the “UP/LAND” configuration.

1) Remove fuselage to landing gear leg fairings (refer to 32-10-11, Main Gear Leg – General Data), and using main inboard jacking points and nose jacking beam (SD12538), jack aircraft until all wheels are at least 8 inches clear of ground (refer to 07-00-00, Lifting and Shoring – General Data).

2) Using a board to protect plastic bottom surface of ski, jack ski under inboard beam, below axle, sufficiently to take weight of ski.

3) Select POWER SOURCE and MASTER switches to OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches to OFF (Post Mod 6/2314).

4) Disengage HYDRAULIC OIL PUMP circuit breaker and discharge hydraulic accumulator pressure by operating brake pedals.

5) Disconnect ski hydraulic lines and electrical connector from main gear leg. Blank open hydraulic lines.

6) Slacken and disconnect ski raising cable from inboard crank assembly lever. Disconnect ski pitch up trim cable from inboard side of sling.

7) Lower and remove jack from under ski.

8) Disconnect link assembly from upper arm of the axle outboard lug.

9) Disconnect restraining cable, forward limit link, and aft limit cable and bungee cord, from the three fittings on the undersurface of the wing.

10) Support (or chock) ski and disconnect inboard and outboard crank assembly levers from their respective axle lugs, noting position of washers. Lower ski to ground.

11) Slide ski forward past wheel.

12) Remove outboard lug from axle.

13) Remove main wheel assy as per 32-40-51, Main Wheels (Mod 6/2002) – General Data, Para 2.A.

14) Loosen torque plate bolts, rotate torque plate 60° FWD (2 bolt positions) to standard gear configuration on axle.

15) Remove inboard lug and replace lower portion ski hardware with original gear hardware (103–22100 bolts, Qty 12).

16) Install wheel assy as per 32-40-51, Main Wheels (Mod 6/2002) – General Data, Para 2.B.

17) Repeat Para 5.A.2) through Para 5.A.16) for opposite ski and, if necessary, remove nose Wheel-Ski (refer to 32-21-00, Nose Wheel-Ski (Mod S.O.O. 6001) – General Data).

18) Lower aircraft to ground and remove jacks.

19) Switch on electrical power by selecting POWER SOURCE switch to EXTERNAL or BATTERY and the MASTER switch to ON (Pre Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

    Switch on electrical power by selecting EXTERNAL or BATTERY switch to ON (Post Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

20) Engage HYDRAULIC OIL PUMP circuit breaker.

    Note Observe that motor pump charges accumulators, and that motor cuts out when pressure reaches 1575 ± 50 psi.

21) Using a suitable container to receive oil spillage, bleed both main Wheel-Ski hydraulic lines, by slackening blanking caps on ski down lines with ski selector lever selected to DOWN/SNOW. When fluid is free of air, tighten blanking cap while fluid is still flowing.

22) Repeat Para 5.A.21) on ski up lines with an UP/LAND selection made.

23) Ensure on completion that all landing gear leg hose blanks are tight and secure.

24) Remove ski selector valve from flight compartment.

25) Install cover plate with mounting screws (4) on instrument panel.

26) Switch OFF electrical supply. Lock out GEAR IND circuit breaker on avionics circuit breaker panel.

27) Install leg fairings and store Wheel-Skis complete with attachment fittings and hardware for future use.

28) Replenish hydraulic system reservoir as necessary (refer to 12-10-15, Hydraulic System – General Data).

29) Change “wheel-ski” airspeed programming instructions with standard gear airspeed program instructions (refer to 32-10-11, Main Gear Leg– General Data, Para 2.C.).

B. Remove Ski Base (Mod 6/2199)

Refer to Figure 2 for details.

1) Remove the Main Wheel-Ski from aircraft as per Para 5.A.

2) Remove the screws securing the skids and skegs to the bottom of ski.

3) Remove skids and skegs.

4) Remove the screws (3) securing the retainer clip to the bottom of ski.

Note It is not necessary to remove doubler, eye bolt or related hardware for the Main Wheel-Ski forward limit cable attachment point.

5) Remove the screws securing the ski base retainers.

6) Remove retainers from ski base.

7) Remove remaining screws securing the ski base to ski.

8) Remove ski base from ski.

C. Preparation of Replacement Ski Base Components

1) Clamp together the old ski base to the replacement ski base.

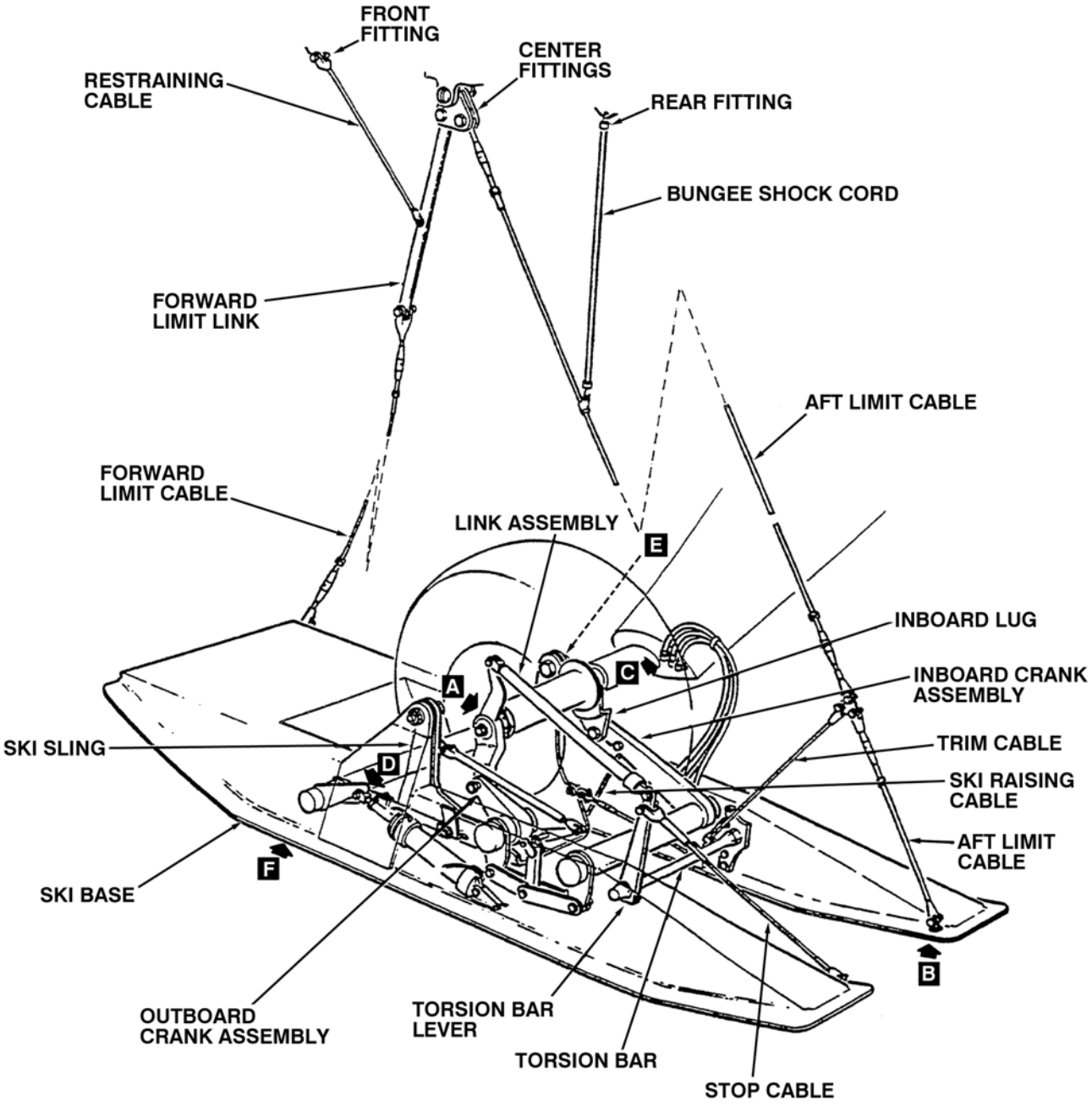
2) Copy profile and hole locations.

3) Transfer hole pattern from ski to replacement ski base. Match hole size and countersunk as required.

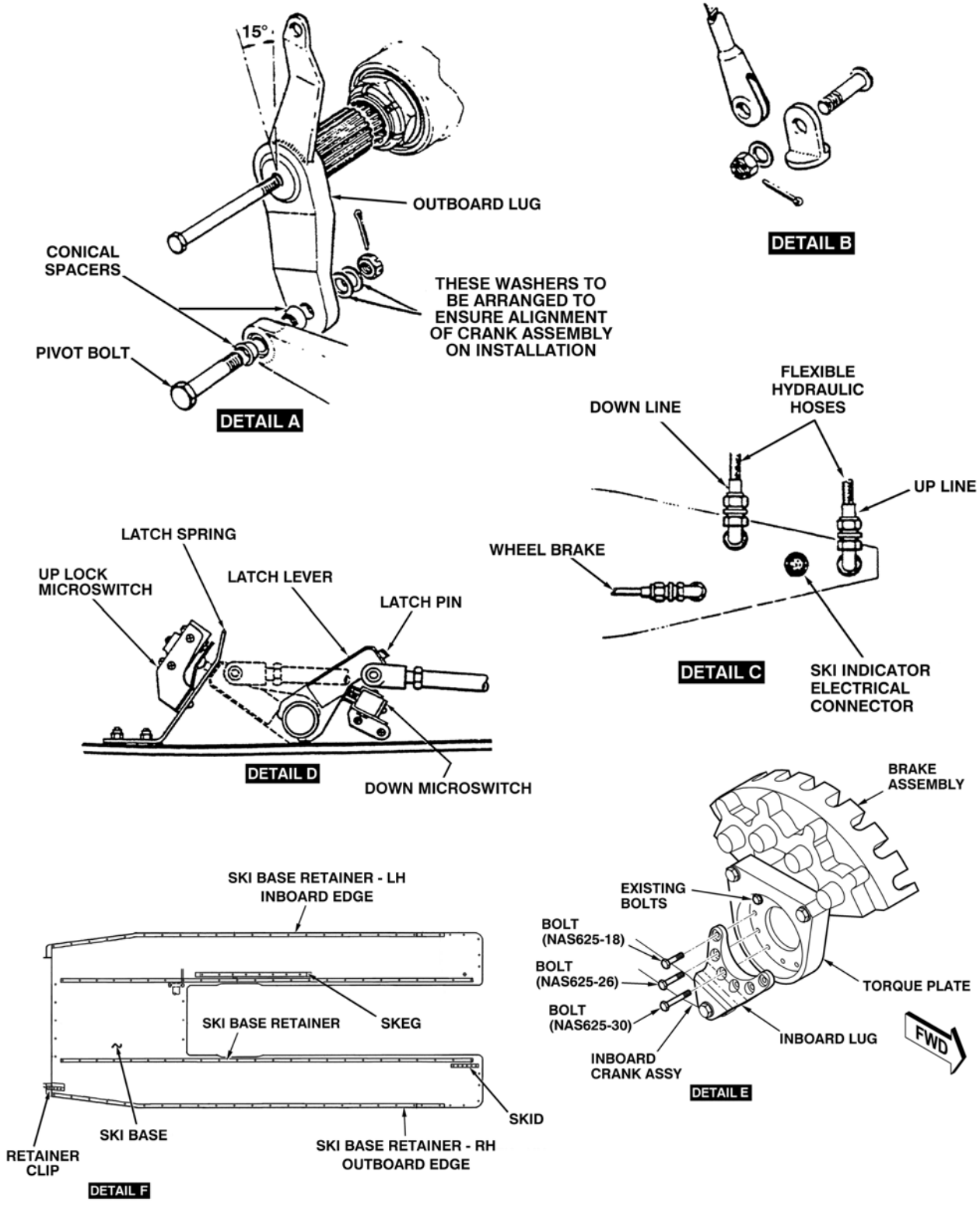
4) Carefully trim replacement base to match old base profile.

5) Using old ski base retainers, bend new retainers to match profile.

6) Clamp retainers securely together and transfer hole patterns. Match hole size and countersunk as required.



Main Wheel-Ski Installation
Figure 2 (Sheet 1 of 2)



Main Wheel-Ski Installation
Figure 2 (Sheet 2 of 2)

D. Install Main Wheel-Skis

Note For the purpose of the following procedures it is assumed that the aircraft has the necessary structural provisions for a Wheel-Ski installation.

1) Select POWER SOURCE and MASTER switches to OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches to OFF (Post Mod 6/2314).

2) Disengage HYDRAULIC OIL PUMP circuit breaker and discharge hydraulic accumulator pressure by operating brake pedals.

3) Remove fuselage to landing gear leg fairing (refer to 32-10-11, Main Gear Leg – General Data) and using main inboard jacking points and nose jacking beam (SD12538), jack aircraft until all wheels are at least 8 inches clear of ground (refer to 07-00-00, Lifting and Shoring – General Data). Level aircraft (refer to 08-00-00, Leveling and Weighing – General Data).

4) Remove main wheel assy as per 32-40-51, Main Wheels (Mod 6/2002) – General Data, Para 2.A.

5) Loosen torque plate bolts, rotate torque plate 60° AFT (2 bolt positions) to clear the ski assembly.

6) Install the inboard lug with the supplied hardware.

Note The wheel-ski installation requires the original torque plate hardware for the upper portion of the torque plate and new hardware for the lower portion, refer to Figure 2, Detail E. (Store the original hardware not used).

7) Install wheel assembly as per 32-40-51, Main Wheels (Mod 6/2002) – General Data, Para 2.B., steps 1 to 4.

8) Reinstall brake assembly as per 32-40-31, Wheel Brakes Unit (Mod 6/2002) – General Data, Para 2.B.

9) Install outboard lug. When installing lug, ensure that centerline of lower arm is 15° to the rear of vertical, before engaging splines. Tighten retaining bolt.

10) Inspect entire ski assembly for corrosion and damage. Ensure ski mechanism is in the “UP/LAND” position (sling up). Connect ski raising cable and leave slack. Check ski moving parts for freedom of movement.

11) Lubricate ski (refer to 12-20-10, Lubrication – General Data, Figure 2).

12) Position ski under main wheel, lift and chock ski until inboard and outboard crank assembly levers align with inboard and outboard axle lug arms (lower arm on outboard lug).

13) Install conical spacers and insert pivot bolts to secure ski inboard and outboard crank assembly levers to axle lugs. Observe the position of the conical spacers, and if necessary, add or interchange washers as required to maintain natural alignment and freedom of movement of the crank assembly levers after installation. Install and tighten pivot bolt nuts and secure with cotter pins.

14) Using a board to protect plastic surface of ski bottom, jack ski under inboard beam, below axle to clear chocks and until the rubber pad on the ski beam contacts landing gear leg. Compression of the rubber pad by 0.125 inch maximum is permissible. Remove chocks.

15) Adjust tension of ski raising cable so that ski will remain up when jack is removed. Remove jack supporting ski.

16) Pitching ski up as required, install link assembly between outboard axle lug upper arm and upper hole in torsion bar lever.

17) Pitching ski down against torsion bar load, install stop cable between eye bolt on outboard ski beam and lower hole in torsion bar lever. Adjust cable as required to bring ski approximately level.

18) Install forward limit link to front hole in center fitting on wing undersurface. Connect restraining cable to wing undersurface forward fitting and center of forward limit link. Connect forward limit cable to forward limit link, but do not at this stage connect to ski.

19) Install aft limit cables to rear hole in center fitting on wing undersurface. Do not at this stage connect to ski.

20) Install hydraulic flexible hoses between landing gear leg and ski inboard beam connections. When tightening, position hoses to prevent any possible contact with wheel.

21) Repeat Para 5.D.9) through Para 5.A.20) to install opposite ski.

22) Switch on electrical power by selecting POWER SOURCE switch to EXTERNAL or BATTERY and the MASTER switch to ON (Pre Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

    Switch on electrical power by selecting EXTERNAL or BATTERY switch to ON (Post Mod 6/2314). Ensure that BUS TIE switch is set to NORMAL.

23) Engage HYDRAULIC OIL PUMP circuit breaker.

    Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi.

24) Select ski selector to DOWN/SNOW and, using a suitable container, bleed both main ski down hydraulic lines by slackening both down lines, at bottom of the landing gear legs, in turn. When fluid is clear of air, retighten each connection while fluid is flowing. Repeat this procedure on both main ski up lines after making an UP/LAND selection on ski selector.

    Note If nose Wheel-Ski is being installed at same time as main Wheel-Skis, bleed nose Wheel-Ski hydraulic lines when carrying out Para 5.D.24) (refer to 32-21-00, Nose Wheel-Ski (Mod S.O.O. 6001) – General Data).

25) With ski selector still selected at UP/LAND, and hydraulic pressure maintained, adjust the stop cable to bring ski level with aircraft datum.

26) Select ski selector to DOWN/SNOW and check level of ski. If a discrepancy of more than 0.5° from level exists, readjust stop cable to average the two.

27) Connect aft limit cables to eye bolt at rear of ski inboard beam. Adjust turnbuckles to take up slack only.

28) Adjust forward limit cable to allow 7 inches of slack and connect to eye bolt at front of ski.

29) Select ski selector to UP/LAND, connect sling trim cable to link joining aft limit cables. Adjust sling cable to take up slack in aft limit cables. Ensure rigged angle of ski is not affected by this adjustment.

30) Measure 53 inches down aft limit cable from center of cable attachment fork end, at wing undersurface center fitting. Secure clamp to cable, connect bungee cord to clamp and rear fitting at wing undersurface.

31) Lock all cable turnbuckles and attachments.

32) Repeat Para 5.D.25) through Para 5.D.31) for opposite main Wheel-Ski.

33) Connect both ski position indicator electrical cables between connectors on inboard ski beams and landing gear legs. Unlock and engage GEAR IND avionics circuit breaker.

34) Function test main and nose Wheel-Skis (refer to 32-30-00, Extension and Retraction (Wheel-Ski) (Mod S.O.O. 6001) – General Data).

35) Test Wheel-Ski indication system (refer to 32-61-00, Wheel-Ski Indication – General Data).

36) Lower aircraft to ground and remove jacks. Install fuselage to leg fairings.

37) Check hydraulic system fluid level (refer to 12-10-15, Hydraulic System – General Data).

38) Change standard gear airspeed program instructions with “wheel-ski” airspeed programming instructions (refer to 32-10-11, Main Gear Leg – General Data, Para 2.B.).

E. Install Ski Base (Mod 6/2199)

1) Ensure bottom of ski is clean and undamaged.

2) Install ski base to ski and secure using new hardware in countersunk holes on base.

Note Over torquing of screws will damage holes in UHMW ski base.

3) Install retainer clip with new hardware.

4) Install the inner ski base retainers. Start at forward end of ski, use new hardware.

5) Install the outer edge ski base retainers. Start at forward end of ski, use new hardware.

6) Install the skids and skegs with associated screws.

6. Adjustment/Test

Note The access panels form part of the stressed structure of the ski and must be installed as soon as adjustments are completed.

A. Adjust Main Ski Attitude Compensating Mechanism

Note Adjustment of compensating mechanism may be required after component replacement and can be done with ski installed or removed from aircraft.

1) If skis are installed jack up aircraft (refer to 07-00-00, Lifting and Shoring – General Data) to take load off skis.

2) Remove outboard aft access panel.

3) Select ski down position.

4) Adjust shaft connecting compensating mechanism to sling (refer to Figure 1) until compensating links are in line with each other. Lock fork end of shaft.

5) Install access panel and, if applicable, lower aircraft.

B. Adjust Sling Actuating Mechanism

1) Jack up aircraft (refer to 07-00-00, Lifting and Shoring – General Data).

2) Remove forward outboard access panels.

3) Select ski UP/LAND position and observe behavior of ski and latch mechanism.

4) If latch pin does not engage locking hole of latch spring, disconnect and adjust fork end of hydraulic actuator until latch pin engages locking hole (refer to Figure 2).

5) If latch pin engages locking hole of latch spring, release hydraulic pressure and check that clearance between tire and sling is between 0.10 inch and 0.60 inch.

Note Tire clearance is adjusted during manufacture and assembly, and should not be affected appreciably in normal service or by the replacement of related components. Ensure that inadequate tire clearance has not been caused by damage to the latch mechanism or ski bottom in area of latch assembly before attempting tire clearance adjustment.

6) Adjust tire clearance by removing or adding shims under base of latch spring (removing shims increases tire clearance).

Note Adjusting fork end of hydraulic actuator will not alter tire clearance and will only lead to erratic or faulty latch function. If, however, shims have been added or removed, adjustment of hydraulic actuator is also required as in Para 6.B.7).

7) Adjust fork end of hydraulic actuator to give 0.015 inch to 0.060 inch overtravel after latch engagement.

8) Check adjustment of Ski Up and Ski Down limit switches. (Refer to 32-61-00, Wheel-Ski Indication – General Data).

9) Reinstall access panels.

10) Lower aircraft.

7. Inspection/Check

A. Check Main Wheel-Ski Rigging

1) Remove fuselage to leg fairings (refer to 32-10-11, Main Gear Leg – General Data), jack aircraft well clear of ground using inboard jacking points and nose jacking beam (SD12538) (refer to 07-00-00, Lifting and Shoring – General Data). Level aircraft (refer to 08-00-00 Leveling and Weighing – General Data).

2) Check tire inflation pressure (refer to 12-10-20, Landing Gear – General Data).

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4) Engage GEAR IND and HYDRAULIC OIL PUMP circuit breakers.

Note Observe that motor pump charges accumulators and that motor cuts out when pressure reaches 1575 ± 50 psi.

5) Operate skis UP/LAND and DOWN/SNOW several times, and check ski mechanical linkage for freedom of movement.

6) Select ski selector lever to UP/LAND and check that main skis are level with aircraft datum and that rubber pad on ski contacts landing gear leg.

Note If ski does not contact leg, adjust ski raising cable (refer to Para 5.D).

7) Select ski selector lever to DOWN/SNOW and check main skis are level with aircraft datum.

Note If a discrepancy of more than 1/2° from level exists as checked in Para 7.A.6) and Para 7.A.7), readjust the stop cable to average the two.

8) Check aft limit cables are adjusted so that only cable slack is taken up.

9) Check forward limit cable has 7 inches of slack to accommodate additional ski down travel.

10) Check that sling trim cables are adjusted to take up slack in aft limit cables. Ensure rigged angle of ski is not affected by this adjustment.

11) Check all cable turnbuckles are in safety and that they are lock pinned or wire locked.

12) Lower aircraft to ground and remove jacking equipment. Install fuselage to leg fairings.

13) Check hydraulic system fluid level (refer to 12-10-15, Hydraulic System – General Data).

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).